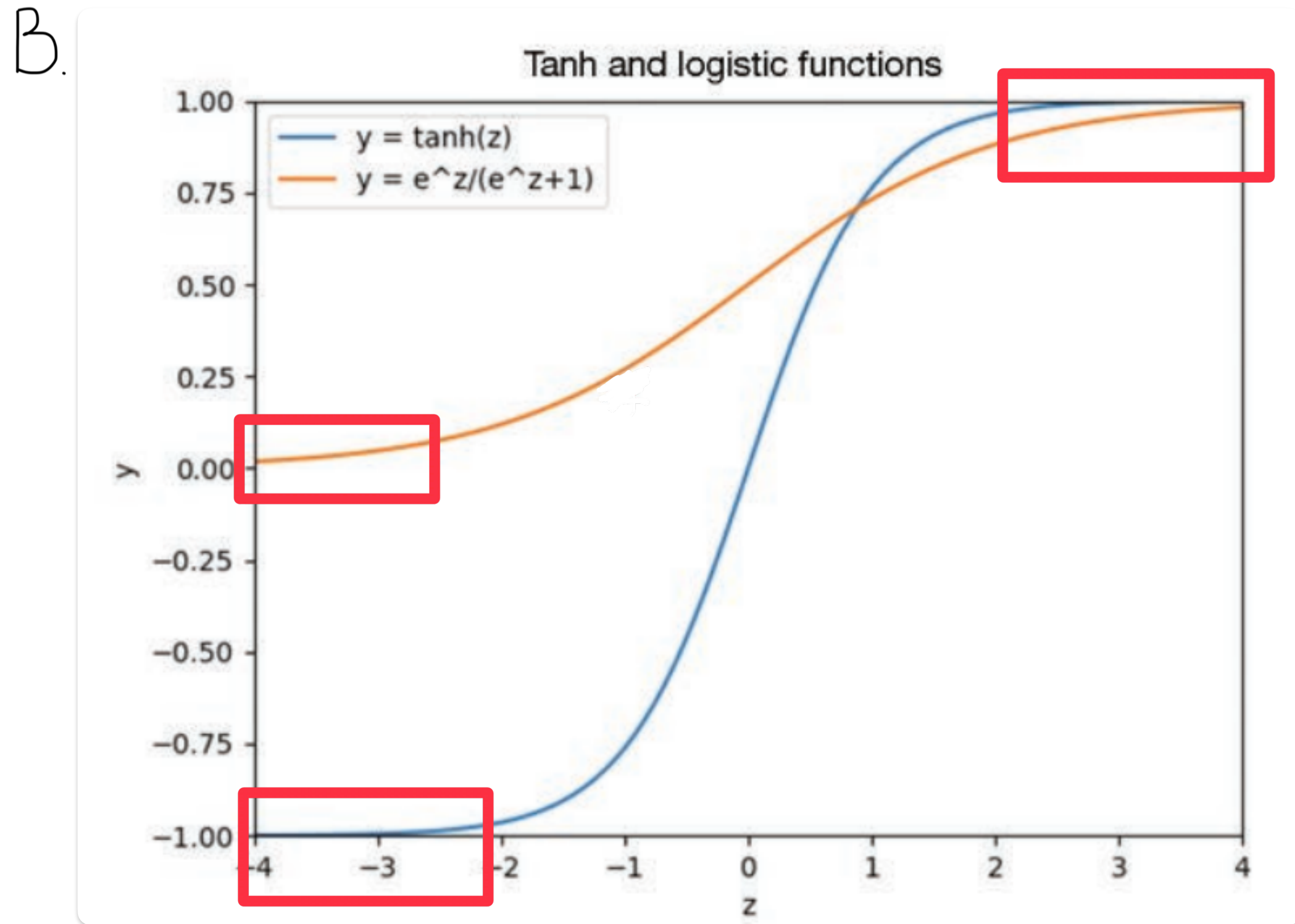


1. Concept of Saturated Neuron:

A. observing one of activation functions such as the hyperbolic tangent function or the sigmoid function on the graph below, we can find out when the input value of an activation function ($\equiv$ the value of weighted sum function) is **very large** or **very small**, the curve of activation function will approach a **horizontal straight line** ($\equiv$ derivative of activation function approach 0), at this point, any change of the input value of an activation function will not affect the output value of an activation function. we call the neuron in this state as a **saturated neuron**.



2. Concept of Vanishing Gradient:

because the backpropagation algorithm requires computing derivative of activation function to receive value of gradient descent, when a neuron saturates, its derivative of activation function approaches 0, and value of gradient descent also approaches 0, even a neuron is not fully saturated, as long as derivative of activation function is less than 1 in each layer, value of gradient descent will rapidly decay to near 0 after multiple layers of computation.

this prevents errors from propagating backwards, causing learning processes to stall and weights to stop updating. this is one of the main reasons for the vanishing gradient issue.